

Chapter 11: Pharmacology Question Bank

Complete NBDHE Board Exam Review Questions & Clinical Rationales

Q1. Which of the following drugs is used to treat trigeminal neuralgia?

- a. Phenytoin
- b. Topiramate
- c. Carbamazepine
- d. Lamotrigine

Correct Answer: C. Carbamazepine*

Clinical Rationale: Carbamazepine (Tegretol), an anticonvulsant agent, is the gold standard drug of choice for managing neuropathic pain associated with trigeminal neuralgia (tic douloureux). Phenytoin (A), Topiramate (B), and Lamotrigine (D) are anticonvulsants primarily indicated for seizure disorders, migraine prophylaxis, or bipolar disorder, but carbamazepine remains the established first-line pharmacotherapy for trigeminal neuralgia.

■ **Quality & Clinical Verification Note:** Grammar typo in original text corrected ("Carbamazepine, and antiseizure drug" -"an antiseizure drug").

Q2. All the drugs listed below can cause xerostomia EXCEPT one. What is the EXCEPTION?

- a. Amitriptyline
- b. Fluoxetine
- c. Thioridazine
- d. Nortriptyline

Correct Answer: B. Fluoxetine*

Clinical Rationale: Fluoxetine (Prozac) is a Selective Serotonin Reuptake Inhibitor (SSRI). Unlike tricyclic antidepressants and typical antipsychotics, SSRIs lack strong muscarinic cholinergic receptor blockade and have significantly lower potential to cause severe anticholinergic xerostomia. In contrast, Amitriptyline (A) and Nortriptyline (D) are tricyclic antidepressants (TCAs), and Thioridazine (C) is a first-generation phenothiazine antipsychotic, all of which exhibit potent anticholinergic activity and frequently cause marked dry mouth.

■ **Quality & Clinical Verification Note:** CRITICAL RATIONALE CORRECTION: The original rationale incorrectly classified Thioridazine as a "tricyclic antidepressant". Thioridazine is actually a first-generation phenothiazine antipsychotic with strong anticholinergic properties.

Q3. Pregnant patients should not take:

- a. Amoxicillin
- b. Erythromycin
- c. Penicillin
- d. Tetracycline

Correct Answer: D. Tetracycline*

Clinical Rationale: Tetracycline is strictly contraindicated in pregnancy (FDA Pregnancy Category D). Tetracyclines cross the placenta and chelate with calcium orthophosphate, depositing directly into calcifying fetal bones and unerupted primary and permanent tooth germ matrices. This causes permanent intrinsic yellow-gray-brown discoloration and enamel hypoplasia in the fetus. Amoxicillin (A), Erythromycin base/stearate (B), and Penicillin VK (C) are safe to administer during pregnancy (Category B). Note: Erythromycin estolate is contraindicated in pregnancy due to maternal cholestatic hepatitis risk.

■ **Quality & Clinical Verification Note:** CRITICAL RATIONALE CORRECTION: The original rationale stated that tetracycline "sits in the gingival crevicular fluid of unerupted teeth in the fetus." Unerupted fetal tooth buds have no gingival crevicular fluid (GCF occurs only in the gingival sulcus of erupted teeth); tetracycline incorporates directly into calcifying enamel and dentin matrices.

Q4. Which antiseizure drug is available as a chewable dose form designed for pediatric patients?

- a. Carbamazepine
- b. Topiramate
- c. Levetiracetam
- d. Gabapentin

Correct Answer: A. Carbamazepine*

Clinical Rationale: Carbamazepine (Tegretol) is formulated as a chewable tablet specifically for pediatric seizure management. Dental Board Alert: Chewable carbamazepine contains approximately 63% sucrose, placing pediatric patients at substantial risk for rampant dental caries; parents and children must be counseled to rinse thoroughly or brush immediately after chewing.

■ **Quality & Clinical Verification Note:** Board Exam Enrichment: Added high-yield note regarding the 63% sucrose content and increased caries risk in pediatric patients taking chewable carbamazepine.

Q5. Patients taking metronidazole should avoid:

- a. Mouth rinse without alcohol
- b. Caffeine-free beverages
- c. Mouth rinse with alcohol
- d. Sugar-free beverages

Correct Answer: C. Mouth rinse with alcohol*

Clinical Rationale: Metronidazole (Flagyl) inhibits the enzyme aldehyde dehydrogenase, leading to accumulation of toxic acetaldehyde when alcohol is ingested. This triggers a severe disulfiram-like (Antabuse-like) reaction characterized by violent nausea, vomiting, abdominal cramps, throbbing headache, flushing, and tachycardia. Patients must strictly avoid all alcohol-containing products, including mouth rinses containing alcohol and elixirs, during therapy and for at least 48 to 72 hours post-treatment.

Q6. Which of the following foods can cause significant drug:food interactions?

- a. Orange juice
- b. Grapefruit juice
- c. Avocados
- d. Apple juice

Correct Answer: B. Grapefruit juice*

Clinical Rationale: Grapefruit juice contains furanocoumarins that irreversibly inhibit intestinal Cytochrome P450 3A4 (CYP3A4) enzymes. This significantly diminishes first-pass metabolism, leading to dangerously elevated serum concentrations of CYP3A4 substrate drugs such as calcium channel blockers (nifedipine), statins (lovastatin, atorvastatin, simvastatin), and benzodiazepines (triazolam, midazolam).

Q7. Which drug is associated with photosensitivity?

- a. Amoxicillin
- b. Phenytoin
- c. Aspirin
- d. Tetracycline

Correct Answer: D. Tetracycline*

Clinical Rationale: Tetracyclines (including doxycycline and minocycline) absorb ultraviolet (UV) radiation and generate reactive oxygen species in the skin, causing marked cutaneous phototoxicity and exaggerated sunburn reactions upon minimal sunlight exposure. Patients taking tetracyclines should be counseled to wear sunscreen and protective clothing.

Q8. Which drug is a common angiotensin-converting enzyme inhibitor?

- a. Propranolol
- b. Losartan
- c. Enalapril
- d. Nifedipine

Correct Answer: C. Enalapril*

Clinical Rationale: Enalapril (Vasotec) is an Angiotensin-Converting Enzyme (ACE) inhibitor (generic drug names ending in "-pril"). ACE inhibitors block the conversion of angiotensin I to the potent vasoconstrictor angiotensin II and prevent bradykinin degradation (which can lead to a dry, nonproductive hacking cough in ~10-20% of patients). Propranolol (A) is a nonselective beta-blocker, Losartan (B) is an Angiotensin Receptor Blocker (ARB), and Nifedipine (D) is a calcium channel blocker.

Q9. Aripiprazole (Abilify) is used to treat all the following EXCEPT one. What is the EXCEPTION?

- a. Schizophrenia
- b. Treatment-resistant depression
- c. Anxiety
- d. Bipolar disorder

Correct Answer: C. Anxiety*

Clinical Rationale: Aripiprazole (Abilify) is an atypical (second-generation) antipsychotic functioning as a partial D2 dopamine and 5-HT1A serotonin agonist. It is FDA-approved for the management of schizophrenia (A), adjunctive therapy for major depressive disorder / treatment-resistant depression (B), and bipolar I disorder / acute mania (D). It is not an FDA-approved indication for generalized anxiety disorder.

Q10. Which drug is a common HMG-CoA reductase inhibitor?

- a. Ezetimibe
- b. Lovastatin
- c. Alirocumab
- d. Colestipol

Correct Answer: B. Lovastatin*

Clinical Rationale: Lovastatin (Mevacor) is an HMG-CoA reductase inhibitor ("statin") that competitively inhibits the rate-limiting enzyme in hepatic cholesterol biosynthesis. Ezetimibe (A) is a cholesterol absorption inhibitor. Alirocumab (C) is a PCSK9 inhibitor monoclonal antibody. Colestipol (D) is a bile acid sequestrant.

Q11. Which drug is a common calcium channel blocker?

- a. Enalapril
- b. Nifedipine
- c. Candesartan
- d. Amiloride

Correct Answer: B. Nifedipine*

Clinical Rationale: Nifedipine (Procardia) is a dihydropyridine calcium channel blocker (CCB). High-yield board alert: CCBs (nifedipine, amlodipine, diltiazem, verapamil) inhibit calcium influx into vascular smooth muscle and cardiac cells, but frequently induce drug-induced gingival enlargement (hyperplasia). Enalapril (A) is an ACE inhibitor, Candesartan (C) is an ARB, and Amiloride (D) is a potassium-sparing diuretic.

Q12. The combination of clarithromycin/amoxicillin or metronidazole/esomeprazole is used to treat:

- a. GERD
- b. Treatment-resistant strep throat
- c. Peptic ulcer disease
- d. Treatment-resistant pneumonia

Correct Answer: C. Peptic ulcer disease*

Clinical Rationale: Peptic ulcer disease caused by *Helicobacter pylori* bacterial infection is treated with combination triple therapy: a proton pump inhibitor (such as esomeprazole or omeprazole) combined with two antibiotics (clarithromycin plus amoxicillin, or metronidazole in penicillin-allergic patients) to eradicate the bacteria and promote mucosal healing.

Q13. Which drug is a second-generation antipsychotic drug?

- a. Fluoxetine
- b. Promethazine
- c. Fluphenazine
- d. Risperidone

Correct Answer: D. Risperidone*

Clinical Rationale: Risperidone (Risperdal) is a second-generation (atypical) antipsychotic drug that antagonizes both serotonin (5-HT_{2A}) and dopamine (D₂) receptors, causing fewer extrapyramidal symptoms than first-generation drugs. Fluoxetine (A) is an SSRI antidepressant. Promethazine (B) is a phenothiazine antihistamine/antiemetic. Fluphenazine (C) is a first-generation (typical) high-potency phenothiazine antipsychotic.

Q14. All the following are side effects of antipsychotic drugs EXCEPT one. What is the EXCEPTION?

- a. Acute dystonic reactions
- b. Sialorrhea
- c. Akathisia
- d. Parkinsonian symptoms

Correct Answer: B. Sialorrhea*

Clinical Rationale: Antipsychotic medications block dopamine receptors in the basal ganglia, leading to Extrapyramidal Symptoms (EPS) such as acute dystonia (spasms of tongue, face, and neck), akathisia (motor restlessness), and pseudoparkinsonism (tremors, rigidity). Due to concurrent muscarinic anticholinergic receptor blockade, antipsychotics cause xerostomia (dry mouth), NOT sialorrhea (excessive salivation).

Q15. Patients taking calcium channel blocking drugs are at risk for developing:

- a. Xerostomia
- b. Gingival hyperplasia
- c. Glossitis
- d. Sedation

Correct Answer: B. Gingival hyperplasia*

Clinical Rationale: Gingival hyperplasia (drug-induced gingival enlargement) is a well-documented adverse effect of calcium channel blockers (such as nifedipine, amlodipine, and verapamil). The enlargement begins in the interdental papillae and can cover the crowns of teeth if oral hygiene is poor. The other two major drug classes causing gingival hyperplasia are anticonvulsants (phenytoin) and immunosuppressants (cyclosporine).

Q16. Patients taking daily aspirin therapy to minimize the risk of an additional myocardial infarction or stroke can experience:

- a. Increased bleeding during scaling and planing
- b. Decreased bleeding during scaling and planing
- c. Xerostomia
- d. Sialorrhea

Correct Answer: A. Increased bleeding during scaling and planing*

Clinical Rationale: Aspirin (acetylsalicylic acid) irreversibly inhibits platelet cyclooxygenase-1 (COX-1), preventing the synthesis of thromboxane A₂ (TXA₂) for the entire lifespan of the platelet (7-10 days). This irreversibly impairs platelet aggregation and prolongs bleeding time, manifesting as increased bleeding during dental scaling and root planing.

Q17. Which drug is a common beta-blocker drug?

- a. Enalapril
- b. Spironolactone
- c. Metoprolol
- d. Prazosin

Correct Answer: C. Metoprolol*

Clinical Rationale: Metoprolol (Lopressor, Toprol-XL) is a cardioselective beta-1 adrenergic receptor blocker (generic names ending in "-olol"). Enalapril (A) is an ACE inhibitor, Spironolactone (B) is an aldosterone antagonist potassium-sparing diuretic, and Prazosin (D) is an alpha-1 blocker.

Q18. All the following drugs are used to alter blood coagulation EXCEPT one. What is the EXCEPTION?

- a. Warfarin
- b. Fenofibrate
- c. Rivaroxaban
- d. Apixaban

Correct Answer: B. Fenofibrate*

Clinical Rationale: Fenofibrate (Tricor) is a fibric acid derivative used to lower blood triglyceride and LDL levels; it does not alter the coagulation cascade or platelet function. Warfarin (Coumadin, A) is a vitamin K antagonist, and Rivaroxaban (Xarelto, C) and Apixaban (Eliquis, D) are direct Factor Xa inhibitor anticoagulants.

Q19. Which drug is used as a rescue inhaler for acute asthma symptoms?

- a. Albuterol
- b. Salmeterol
- c. Cromolyn
- d. Flunisolide

Correct Answer: A. Albuterol*

Clinical Rationale: Albuterol (Ventolin, ProAir) is a Short-Acting Beta-2 Agonist (SABA) with rapid onset of bronchodilation (within 5-15 minutes), making it the first-line emergency rescue inhaler for acute bronchospasm. Salmeterol (B) is a Long-Acting Beta-2 Agonist (LABA) with a slow onset used strictly for maintenance. Cromolyn (C) is a mast-cell stabilizer for long-term prophylaxis. Flunisolide (D) is an inhaled corticosteroid for daily maintenance.

Q20. Which drug can be safely used in a pregnant woman?

- a. Tetracycline
- b. Alprazolam
- c. Amoxicillin
- d. Diazepam

Correct Answer: C. Amoxicillin*

Clinical Rationale: Amoxicillin is an FDA Pregnancy Category B antibiotic and is the primary antibiotic of choice for dental infections in pregnant patients. Tetracycline (A) is Category D (causes permanent intrinsic tooth discoloration and bone growth inhibition). Alprazolam (B) and Diazepam (D) are benzodiazepines (Category D/X) associated with teratogenic risks (cleft palate) and neonatal floppy infant syndrome.

Q21. Patients taking zolpidem (Ambien) for insomnia can experience all the following EXCEPT one. What is the EXCEPTION?

- a. Sleep-driving
- b. Sleep-eating
- c. Agitation
- d. Next day sedation

Correct Answer: C. Agitation*

Clinical Rationale: Zolpidem (Ambien) is a non-benzodiazepine hypnotic that binds selectively to the alpha-1 subunit of GABA-A receptors. Known adverse effects include complex parasomnia sleep-related behaviors such as sleep-driving (A), sleep-eating (B), sleep-walking, and next-day residual sedation/grogginess (D). Agitation is not a standard side effect.

Q22. Which drug irreversibly binds to the proton pump, resulting in acid suppression that lasts more than 24 hours?

- a. Famotidine
- b. Esomeprazole
- c. Sucralfate
- d. Magnesium

Correct Answer: B. Esomeprazole*

Clinical Rationale: Esomeprazole (Nexium) is a Proton Pump Inhibitor (PPI) that forms covalent disulfide bonds with the H⁺/K⁺ ATPase enzyme system in gastric parietal cells, causing irreversible enzyme inactivation and gastric acid suppression lasting >24 hours until new pumps are synthesized. Famotidine (A) is a reversible H₂-blocker. Sucralfate (C) forms a protective paste over ulcers. Magnesium (D) is an inorganic antacid.

Q23. Which drug exerts its antidepressant effect by solely blocking the reuptake of serotonin?

- a. Fluoxetine
- b. Venlafaxine
- c. Duloxetine
- d. Amitriptyline

Correct Answer: A. Fluoxetine*

Clinical Rationale: Fluoxetine (Prozac) is a Selective Serotonin Reuptake Inhibitor (SSRI) that selectively blocks the serotonin transporter (SERT), solely inhibiting serotonin reuptake without significant norepinephrine uptake inhibition. Venlafaxine (Effexor, B) and Duloxetine (Cymbalta, C) are Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs). Amitriptyline (Elavil, D) is a nonselective tricyclic antidepressant (TCA).

Q24. Which drug blocks histamine2-receptors in the stomach?

- a. Esomeprazole
- b. Magnesium
- c. Famotidine
- d. Aluminum

Correct Answer: C. Famotidine*

Clinical Rationale: Famotidine (Pepcid) is an H₂-receptor antagonist (ending in "-tidine") that competitively blocks histamine₂-receptors on gastric parietal cells, decreasing baseline and food-stimulated hydrochloric acid secretion. Esomeprazole (A) is a proton pump inhibitor. Magnesium (B) and Aluminum (D) are neutralizing antacid bases.

Q25. Which drug is used to treat opioid overdose?

- a. Flumazenil
- b. Epinephrine
- c. Diphenhydramine
- d. Naloxone

Correct Answer: D. Naloxone*

Clinical Rationale: Naloxone (Narcan) is a pure, competitive opioid receptor antagonist (μ , κ , δ) that rapidly reverses life-threatening opioid-induced respiratory depression, central nervous system depression, and miosis. Flumazenil (Romazicon, A) is a benzodiazepine antagonist. Epinephrine (B) is used for anaphylaxis and cardiac arrest. Diphenhydramine (Benadryl, C) is an H₁-antihistamine.

■ **Quality & Clinical Verification Note:** MISSING ANSWER IN ORIGINAL DOCUMENT: Solved and provided complete clinical rationale.

Q26. Patients taking large doses of acetaminophen for pain should be warned of the risk of:

- a. Sedation
- b. Liver toxicity
- c. Gastric bleeding
- d. Ototoxicity

Correct Answer: B. Liver toxicity*

Clinical Rationale: Acetaminophen (Tylenol / APAP) hepatotoxicity occurs when excessive doses (>3000–4000 mg/day, or lower in chronic alcohol users) deplete glutathione, leading to accumulation of the toxic reactive intermediate NAPQI (N-acetyl-p-benzoquinone imine), which binds covalent bonds to hepatocytes causing acute liver necrosis. Gastric bleeding (C) is a risk of NSAIDs and aspirin, not acetaminophen. Sedation (A) occurs with opioids. Ototoxicity (D) is associated with aminoglycosides and high-dose aspirin.

■ **Quality & Clinical Verification Note:** MISSING ANSWER IN ORIGINAL DOCUMENT: Solved and provided complete clinical rationale.

Q27. Patients with renal impairment should avoid the following drugs EXCEPT for one. What is the EXCEPTION?

- a. Ibuprofen
- b. Acetaminophen
- c. Naproxen
- d. Meloxicam

Correct Answer: B. Acetaminophen*

Clinical Rationale: Acetaminophen is primarily metabolized by hepatic glucuronidation and sulfation and lacks renal prostaglandin inhibitory effects, making it the safest non-opioid analgesic in patients with chronic kidney disease or renal failure. In contrast, NSAIDs (Ibuprofen [A], Naproxen [C], Meloxicam [D]) inhibit renal prostaglandin (PGE₂, PGI₂) synthesis, causing renal vasoconstriction, acute reduction in GFR, and risk of acute tubular necrosis.

■ **Quality & Clinical Verification Note:** MISSING ANSWER IN ORIGINAL DOCUMENT: Solved and provided complete clinical rationale.

Q28. Patients using nicotine chewing gum to stop smoking should be counseled about all the following side effects EXCEPT one. What is the EXCEPTION?

- a. Jaw ache
- b. The ability of the gum to stick to dental work
- c. Runny nose
- d. Sore mouth

Correct Answer: C. Runny nose*

Clinical Rationale: Runny nose (rhinorrhea) and nasal irritation are characteristic side effects of nicotine nasal spray, not nicotine gum. Nicotine gum side effects include jaw soreness/ache (A, from improper continuous chewing rather than the recommended "chew and park" technique), sticking to dental restorations/crowns/dentures (B), and sore mouth/throat irritation (D).

■ **Quality & Clinical Verification Note:** MISSING ANSWER IN ORIGINAL DOCUMENT: Solved and provided complete clinical rationale.

Q29. All the following are side effects of varenicline EXCEPT one. What is the EXCEPTION?

- a. Changes in mood and behavior
- b. Sleep problems
- c. Constipation
- d. Diarrhea

Correct Answer: D. Diarrhea*

Clinical Rationale: Varenicline (Chantix) is an alpha-4 beta-2 nicotinic acetylcholine receptor partial agonist for smoking cessation. Documented side effects include neuropsychiatric changes (mood alterations, depression, abnormal/vivid dreams [A, B]), nausea (~30%), flatulence, and constipation (C). Diarrhea is not a characteristic adverse effect of varenicline.

■ **Quality & Clinical Verification Note:** MISSING ANSWER IN ORIGINAL DOCUMENT: Solved and provided complete clinical rationale.

Q30. Which drug is used to treat dental anxiety?

- a. Zolpidem
- b. Alprazolam
- c. Butabarbital
- d. Buspirone

Correct Answer: B. Alprazolam*

Clinical Rationale: Alprazolam (Xanax) is a short-to-intermediate acting benzodiazepine commonly prescribed for oral pre-procedural anxiolysis and conscious sedation in patients with acute dental anxiety. Zolpidem (A) is a hypnotic indicated for insomnia. Butabarbital (C) is an intermediate-acting barbiturate with high abuse and respiratory depression liability. Buspirone (D) requires 2 to 4 weeks of continuous daily administration and cannot be used PRN for acute situational dental anxiety.

■ **Quality & Clinical Verification Note:** MISSING ANSWER IN ORIGINAL DOCUMENT: Solved and provided complete clinical rationale.

Clinical Rationale: ## ■ CLINICAL CASE A: ADULT PERIODONTAL & MEDICAL REVIEW

Clinical Rationale: ### Patient Profile: Female, 65 years old

Clinical Rationale: * Chief Complaint: Routine oral health examination

Clinical Rationale: * Medical History: Type II diabetes, Hypertension, Hypercholesterolemia

Clinical Rationale: * Current Medications: Metformin, Semaglutide (Ozempic) SubQ, Enalapril/Hydrochlorothiazide, Atorvastatin (Lipitor)

Clinical Rationale: * Clinical Findings: BP 130/80 mmHg, A1C 7.0% (3 months ago)

Q31 (Case A.1). Why is the patient advised to eat potassium-rich foods such as bananas or orange juice while taking hydrochlorothiazide (HCTZ)?

- a. To increase sodium retention
- b. To accelerate the rate of diuresis
- c. To replenish potassium lost due to thiazide diuretic-induced hypokalemia
- d. To enhance gastrointestinal drug absorption

Correct Answer:*C. To replenish potassium lost due to thiazide diuretic-induced hypokalemia**

Clinical Rationale: Thiazide diuretics (such as hydrochlorothiazide) inhibit sodium and chloride reabsorption in the distal convoluted tubule, causing potassium wasting and hypokalemia. Dietary potassium from orange juice or bananas helps replenish serum potassium.

Q32 (Case A.2). Angiotensin-converting enzyme (ACE) inhibitors tend to increase serum potassium levels, whereas thiazide diuretics increase potassium excretion.

- a. Both statements are TRUE.
- b. Both statements are FALSE.
- c. The first statement is TRUE; the second statement is FALSE.
- d. The first statement is FALSE; the second statement is TRUE.

Correct Answer:*A. Both statements are TRUE.**

Clinical Rationale: Both statements are true. ACE inhibitors (like enalapril) suppress aldosterone, causing potassium retention (hyperkalemia risk), while thiazide diuretics increase potassium excretion (hypokalemia risk). Combined therapy helps maintain balanced potassium levels.

Q33 (Case A.3). To prevent orthostatic hypotension in patients taking antihypertensive medications like enalapril/HCTZ, the dental hygienist should:

- a. Dismiss the patient immediately after uprighting the chair
- b. Raise the dental chair slowly and have the patient sit upright for several minutes before standing
- c. Administer 100% oxygen prior to standing
- d. Place the patient in Trendelenburg position before standing

Correct Answer:*B. Raise the dental chair slowly and have the patient sit upright for several minutes before standing**

Clinical Rationale: Antihypertensives cause postural (orthostatic) hypotension due to altered sympathetic tone. Raising the chair slowly and allowing the patient to dangle their legs and establish circulation prevents syncope and lightheadedness.

Q34 (Case A.4). Which medication taken by the patient functions as an HMG-CoA reductase inhibitor to lower serum cholesterol?

- a. Atorvastatin (Lipitor)
- b. Niacin
- c. Gemfibrozil
- d. Cholestyramine

Correct Answer:*A. Atorvastatin (Lipitor)**

Clinical Rationale: Atorvastatin (Lipitor) is an HMG-CoA reductase inhibitor ("statin") that competitively inhibits the rate-limiting enzyme in hepatic cholesterol biosynthesis.

Q35 (Case A.5). Which cholesterol-lowering medication works by binding to bile acids in the intestine to form an insoluble complex excreted in feces?

- a. Clofibrate
- b. Ezetimibe
- c. Icosapent ethyl
- d. Cholestyramine

Correct Answer: D. Cholestyramine*

Clinical Rationale: Cholestyramine is a bile acid sequestrant resin that binds bile acids in the intestinal lumen, preventing their reabsorption and forcing the liver to utilize circulating LDL cholesterol to synthesize new bile acids.

Q36 (Case A.6). In patients with Type 2 diabetes and hypertension, ACE inhibitors (such as enalapril) are considered first-line agents primarily because they:

- a. Directly stimulate insulin release from pancreatic beta cells
- b. Eliminate the risk of hypoglycemia
- c. Retard the progression of diabetic nephropathy and protect renal microvasculature
- d. Prevent all forms of diabetic neuropathy

Correct Answer: C. Retard the progression of diabetic nephropathy and protect renal microvasculature*

Clinical Rationale: ACE inhibitors dilate the efferent arterioles of the glomerulus, reducing intraglomerular hydrostatic pressure and slowing the progression of diabetic nephropathy, making them the preferred antihypertensive class in diabetic patients.

Q37 (Case A.7). Which of the following newer cholesterol-lowering agents is a monoclonal antibody that functions as a PCSK9 inhibitor?

- a. Ticagrelor
- b. Rivaroxaban
- c. Apixaban
- d. Evolocumab (Repatha)

Correct Answer: D. Evolocumab (Repatha)*

Clinical Rationale: Evolocumab (Repatha) is a human monoclonal antibody that binds to PCSK9, preventing LDL receptor degradation and substantially lowering circulating LDL cholesterol levels.

Q38 (Case A.8). The biguanide oral antihyperglycemic medication metformin carries a boxed warning for which rare but potentially fatal metabolic complication?

- a. Diabetic ketoacidosis
- b. Lactic acidosis
- c. Acute pancreatitis
- d. Thyroid C-cell carcinoma

Correct Answer: B. Lactic acidosis*

Clinical Rationale: Metformin carries a black box warning for lactic acidosis, which occurs when lactic acid accumulates due to impaired hepatic gluconeogenesis and lactate clearance, especially in patients with renal impairment or chronic alcoholism.

Q39 (Case A.9). Which protocol is recommended when scheduling dental hygiene appointments for a diabetic patient taking oral antidiabetic medications?

- a. Schedule morning appointments shortly after the patient has eaten a normal meal and taken their medication
- b. Instruct the patient to fast for 8 hours before the dental hygiene appointment
- c. Require the patient to bring their laboratory A1C test kit to every appointment
- d. Withhold all oral antidiabetic medications on the morning of dental scaling

Correct Answer:*A. Schedule morning appointments shortly after the patient has eaten a normal meal and taken their medication**

Clinical Rationale: Morning appointments scheduled 1.5 to 2 hours after breakfast and morning medication administration provide maximum glycemic stability and reduce the risk of acute hypoglycemic episodes during dental procedures.

Q40 (Case A.10). The patient is taking semaglutide (Ozempic). Semaglutide belongs to which pharmacological class of antidiabetic agents?

- a. Sulfonylurea insulin secretagogue
- b. SGLT2 (Sodium-Glucose Co-transporter 2) inhibitor
- c. GLP-1 (Glucagon-Like Peptide-1) receptor agonist
- d. Thiazolidinedione (TZD) insulin sensitizer

Correct Answer:*C. GLP-1 (Glucagon-Like Peptide-1) receptor agonist**

Clinical Rationale: Semaglutide (Ozempic) is a glucagon-like peptide-1 (GLP-1) receptor agonist that enhances glucose-dependent insulin secretion, slows gastric emptying, and suppresses inappropriate postprandial glucagon secretion.

Clinical Rationale: ## ■ CLINICAL CASE B: GERIATRIC RESPIRATORY & CARDIAC REVIEW

Clinical Rationale: #### Patient Profile: Male, 85 years old

Clinical Rationale: * Chief Complaint: Sore, dry mouth with white milky discharge/plaques

Clinical Rationale: * Medical History: COPD, Seasonal allergies, Atrial fibrillation, Coronary artery disease

Clinical Rationale: * Current Medications: Albuterol inhaler PRN, Breztri Aerosphere (Budesonide/glycopyrrolate/formoterol), Fexofenadine (Allegra), Apixaban (Eliquis), SL Nitroglycerin PRN

Clinical Rationale: * Clinical Findings: BP 125/80 mmHg, Height 6'2", Weight 200 lbs

Q41 (Case B.1). The patient complains of a sore throat and dry mouth. Which component of his inhaler therapy is MOST likely contributing to these oral symptoms?

- a. Inhaled beta-2 adrenergic agonists and anticholinergics administered via oral metered-dose inhaler
- b. Fexofenadine (Allegra)
- c. Apixaban (Eliquis)
- d. Daily multivitamin

Correct Answer:*A. Inhaled beta-2 adrenergic agonists and anticholinergics administered via oral metered-dose inhaler**

Clinical Rationale: Aerosolized inhaled beta-agonists (albuterol/formoterol) and muscarinic anticholinergics (glycopyrrolate) dry and irritate the pharyngeal and oral mucosa, causing xerostomia and pharyngeal irritation.

Q42 (Case B.2). The patient carries an albuterol inhaler. Albuterol is classified as a:

- a. Short-acting bronchodilator used for sudden, acute breathing distress
- b. Long-acting maintenance corticosteroid
- c. Mast-cell stabilizer for seasonal allergy prophylaxis
- d. Leukotriene receptor antagonist

Correct Answer: A. *Short-acting bronchodilator used for sudden, acute breathing distress**

Clinical Rationale: Albuterol (Proventil, Ventolin) is a Short-Acting Beta-2 Agonist (SABA) bronchodilator indicated for rapid emergency relief of acute bronchospasm and dyspnea in COPD and asthma.

Q43 (Case B.3). The patient presents with white milky discharge/plaques in the oral cavity. What is the MOST likely cause of this finding?

- a. Allergic reaction to fexofenadine
- b. Gingival bleeding secondary to apixaban therapy
- c. Vitamin deficiency stomatitis
- d. Oral candidiasis (thrush) secondary to the inhaled corticosteroid component (budesonide)

Correct Answer: D. *Oral candidiasis (thrush) secondary to the inhaled corticosteroid component (budesonide)**

Clinical Rationale: Inhaled corticosteroids (such as budesonide in Breztri) locally suppress mucosal immune resistance in the oral cavity, facilitating the colonization and overgrowth of *Candida albicans* (thrush).

Q44 (Case B.4). What crucial home care instruction should the dental hygienist give this patient to prevent recurrent oral candidiasis from his Breztri inhaler?

- a. Rinse the mouth thoroughly with water and spit it out (or brush) immediately after every inhaler use
- b. Discontinue the inhaler immediately whenever white spots appear
- c. Swallow a full glass of milk before using the inhaler
- d. Use the inhaler only once per week

Correct Answer: A. *Rinse the mouth thoroughly with water and spit it out (or brush) immediately after every inhaler use**

Clinical Rationale: Rinsing the oral cavity vigorously with water and expectorating immediately after using a corticosteroid inhaler eliminates residual drug particles from the mucosal surface, preventing fungal overgrowth.

Q45 (Case B.5). Patients with severe COPD using daily combination inhaled corticosteroids are at an increased risk for which systemic respiratory infection?

- a. Streptococcal pharyngitis
- b. Pulmonary embolism
- c. Pneumonia
- d. Acute rhinitis

Correct Answer: C. *Pneumonia**

Clinical Rationale: Chronic high-dose inhaled corticosteroids suppress local pulmonary alveolar macrophage defenses, increasing the clinical incidence of bacterial pneumonia in elderly patients with COPD.

Q46 (Case B.6). Glycopyrrolate is an active component in the patient's Breztri inhaler. What is the pharmacological mechanism of glycopyrrolate?

- a. Long-acting muscarinic anticholinergic antagonist (LAMA) that blocks acetylcholine receptors on bronchial smooth muscle
- b. Inhaled corticosteroid that inhibits leukotriene synthesis
- c. Short-acting beta-2 adrenergic agonist that stimulates cyclic AMP
- d. Phosphodiesterase-4 (PDE4) inhibitor

Correct Answer: A. Long-acting muscarinic anticholinergic antagonist (LAMA) that blocks acetylcholine receptors on bronchial smooth muscle*

Clinical Rationale: Glycopyrrolate is a Long-Acting Muscarinic Antagonist (LAMA) that competitively blocks acetylcholine at M3 muscarinic receptors in bronchial smooth muscle, preventing cholinergic bronchoconstriction.

Q47 (Case B.7). The patient takes fexofenadine (Allegra) for seasonal allergies. Why does fexofenadine cause significantly less sedation than first-generation antihistamines like diphenhydramine (Benadryl)?

- a. Fexofenadine acts on alpha-1 adrenergic receptors rather than histamine receptors
- b. Fexofenadine is a second-generation H1-blocker that does not readily cross the blood-brain barrier
- c. Fexofenadine stimulates central nervous system serotonin receptors
- d. Fexofenadine undergoes complete first-pass hepatic metabolism

Correct Answer: B. Fexofenadine is a second-generation H1-blocker that does not readily cross the blood-brain barrier*

Clinical Rationale: Fexofenadine is a second-generation peripheral H1-antihistamine. Its polar, lipophobic molecular structure prevents penetration across the blood-brain barrier, resulting in minimal CNS sedation.

Q48 (Case B.8). The patient takes apixaban (Eliquis) for atrial fibrillation. What is the mechanism of action of apixaban?

- a. Vitamin K epoxide reductase inhibitor
- b. Direct thrombin (Factor IIa) inhibitor
- c. Direct Factor Xa inhibitor
- d. Irreversible cyclooxygenase-1 (COX-1) inhibitor

Correct Answer: C. Direct Factor Xa inhibitor*

Clinical Rationale: Apixaban (Eliquis) is a direct-acting oral anticoagulant (DOAC) that selectively and reversibly inhibits Factor Xa in both intrinsic and extrinsic clotting pathways, preventing conversion of prothrombin to thrombin.

Q49 (Case B.9). The patient has a prescription for sublingual nitroglycerin for acute angina. Which protocol should the dental team follow regarding nitroglycerin during dental appointments?

- a. Keep the patient's nitroglycerin bottle on the bracket table and ensure office emergency kit nitroglycerin is accessible
- b. Instruct the patient to take one sublingual tablet prophylactically 30 minutes before every dental visit
- c. Store nitroglycerin in an unsealed plastic cup on the counter
- d. Administer nitroglycerin continuously every 2 minutes if chest pain occurs

Correct Answer: A. Keep the patient's nitroglycerin bottle on the bracket table and ensure office emergency kit nitroglycerin is accessible*

Clinical Rationale: For patients with diagnosed angina, the patient's personal sublingual nitroglycerin must be placed on the bracket table ready for immediate emergency access. Nitroglycerin must remain in its original dark glass bottle to prevent volatilization and plastic adsorption.

Q50 (Case B.10). Nitroglycerin is strictly contraindicated within 24 hours of taking which class of medications due to the risk of severe, fatal hypotension?

- a. Phosphodiesterase-5 (PDE-5) inhibitors (e.g., sildenafil, tadalafil)
- b. Second-generation antihistamines (fexofenadine)
- c. Beta-2 adrenergic agonists (albuterol)
- d. Inhaled corticosteroids (budesonide)

Correct Answer: **A. *Phosphodiesterase-5 (PDE-5) inhibitors (e.g., sildenafil, tadalafil)****

Clinical Rationale: Co-administration of nitrates with PDE-5 inhibitors (such as sildenafil or tadalafil) produces profound, uncontrolled cyclic GMP accumulation, resulting in catastrophic refractory hypotension and cardiovascular collapse.
